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CASE REPORT

PRIMARY BILATERAL ADRENAL LYMPHOMA PRESENTING WITH IMPENDING ADRENAL CRISIS

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Abstract

Primary bilateral adrenal lymphoma is a very rare cause of adrenal insufficiency. We report the case of a 63-year-old woman who presented with signs and symptoms of impending adrenal crisis when referred for evaluation of large bilateral adrenal masses diagnosed on a computed tomography scan two weeks prior. Based on a high clinical suspicion of adrenal insufficiency, patient was initiated on glucocorticoid and mineralocorticoid therapy prior to laboratory confirmation of adrenal insufficiency. After stabilizing the patient and excluding pheochromocytoma, we proceeded with adrenal biopsy that revealed a nongerminal center-type diffuse large B-cell lymphoma. Our patient was treated with R-CHOP chemotherapy, with good response after 3 cycles but eventually died after the fifth cycle from neurologic complications. This case highlights the notion that primary adrenal insufficiency should be considered in patients presenting with bilateral adrenal masses. Although primary adrenal lymphoma is a very rare adrenal malignancy it should be considered in patients presenting with bilateral rapidly growing adrenal tumors and primary adrenal insufficiency.

Keywords: adrenal lymphoma, adrenal crisis, adrenal biopsy.

INTRODUCTION

Primary adrenal lymphoma (PAL) is a very rare adrenal malignancy with a prevalence of 0.2% of all adrenal masses (1) and 2-5% of all adrenal malignancies (2, 3). Bilateral involvement of adrenal glands in patients with primary adrenal lymphoma was reported in up to 30% of cases (2) and may be associated with adrenal insufficiency (4). Once established, primary adrenal insufficiency is usually permanent (5, 6). The most common subtype of PAL is diffuse large B cell lymphoma, with a generally poor prognosis. We report a case of PAL presenting as impending adrenal crisis.

CASE PRESENTATION

A 63-year-old woman was referred to our Endocrinology Department for the evaluation of bilateral large adrenal masses detected on a computed tomography (CT) scan performed two weeks prior to presentation for evaluation of unexplained weight loss and general sense of malaise. She reported symptoms of fatigue, anorexia, nausea, weight loss, and muscle weakness for two months prior to presentation. She noticed improvement in her blood pressure and glucose control without any changes in diet or pharmacotherapy. Her past medical history included hypertension, chronic heart failure, type 2 diabetes mellitus, recurrent atrial fibrillation on chronic anticoagulation (acenocoumarol), and history of total hysterectomy performed for an endometrioma two years prior. On examination, her blood pressure was 100/65 mmHg while she was seated but dropped to 80/60 mmHg after she stood up. Her heart rate was regular at 90 beats per minute, her body mass index was 36.7 kg/m^2, and temperature was 37°C. She appeared to be in distress, was unable to stand without help, and had mild skin hyperpigmentation and moderate peripheral edema. Bilateral flank masses were palpated.

Diagnostic assessment

Laboratory work up revealed hyponatremia, but no hyperkalemia (Table 1). Adrenal hormonal work up was performed due to high clinical suspicion

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for primary adrenal insufficiency and confirmed it with elevated corticotropin (ACTH), inappropriately mid-normal cortisol, and low dehydroepiandrosterone sulfate (DHEAS). Concomitant mineralocorticoid deficiency was diagnosed based on low levels of aldosterone, high renin, and hyponatremia (Table 1).

CT of the abdomen revealed large bilateral adrenal masses measuring 98 ×126 mm on the left and 86 × 118 mm on the right (Fig. 1 A, B) without lymphadenopathy. Both adrenal lesions were solid and heterogeneous, suggestive of hemorrhagic necrosis and demonstrated a moderate post-contrast enhancement. Both kidneys were displaced downwards and could not be easily separated from adjacent structures: the diaphragmatic crus, tail of the pancreas, some segments of the liver; both adrenal masses appeared to encase homolateral renal hilum. The renal vasculature and inferior vena cava were displaced but with normal flux.

After excluding pheochromocytoma, a percutaneous ultrasound-guided core biopsy of the right adrenal mass was performed using a semi-automatic 18-gauge needle.

Histopathology confirmed a diffuse large B-cell lymphoma (DLBCL) type NOS (not otherwise specified) with non-germinal center immunophenotype. Hematoxylin and eosin staining showed nuclear pleomorphism within large malignant lymphoid cells (Fig. 2A), and immunohistochemical staining for the B-cell marker CD20 (Fig. 2B) was positive, as well as for BCL2, but was negative for CD10. Ki-67 was estimated at 90% reflecting a very high mitotic rate (Fig. 2C).

CT of the head, thorax and pelvis revealed no other lesions. A bone marrow biopsy showed no involvement. Additional pertinent laboratory findings included severe anemia (hemoglobin 6.6g/dl, reference range: 12-15.5 g/dl), hypoproteinemia (protein of 5.4 g/dL, reference range: 6.5-8.7 g/dl) and low blood coagulability (INR>15, reference range: 0.85-1.25); APTT 75s, reference range: 22-35 s).

Treatment

The patient was promptly initiated on glucocorticoid therapy with intravenous hydrocortisone that resulted in clinical improvement and normalization

Table 1. Baseline laboratory findings

Laboratory findings	Result	Reference Range
Serum cortisol, µg/dL (nmol/L)	13.82 (381.29)	4.82-19.5 µg/dL (132.96-537.93 nmol/L)
ACTH, pg/mL (pmol/L)	554 (121.99)	3-66 pg/mL (0.66-14.53 pmol/L)
DHEAS, µg/dl (µmol/L)	4.5 (0.12)	12-133 µg/dl (0.32-3.66 µmol/L)
Aldosterone, pg/ml (nmol/L)	22.1 (0.06)	11.7-236 pg/ml (0.03-0.65 nmol/L)
DRC, pg/mL (pmol/L)	>300 (>7.11)	2.64-27.66 pg/mL (0.06-0.65 pmol/L)
Sodium, mmol/L	128	136-145 mmol/L
Potassium, mmol/L	4.6	3.5-5.2 mmol/L
Creatinine µmol/L	70.7	44.2-97.2 µmol/L
Ferritin, µg/L	>2000	4.63-204 µg/L
LDH, U/L (µKat/L)	739 (12.32)	135-214 U/L (2.25-3.57 µKat/L)

Abbreviations: ACTH, adrenocorticotropic hormone; DHEAS, Dehydroepiandrosterone Sulphate; DRC, Direct renin concentration; LDH, Lactate dehydrogenase.

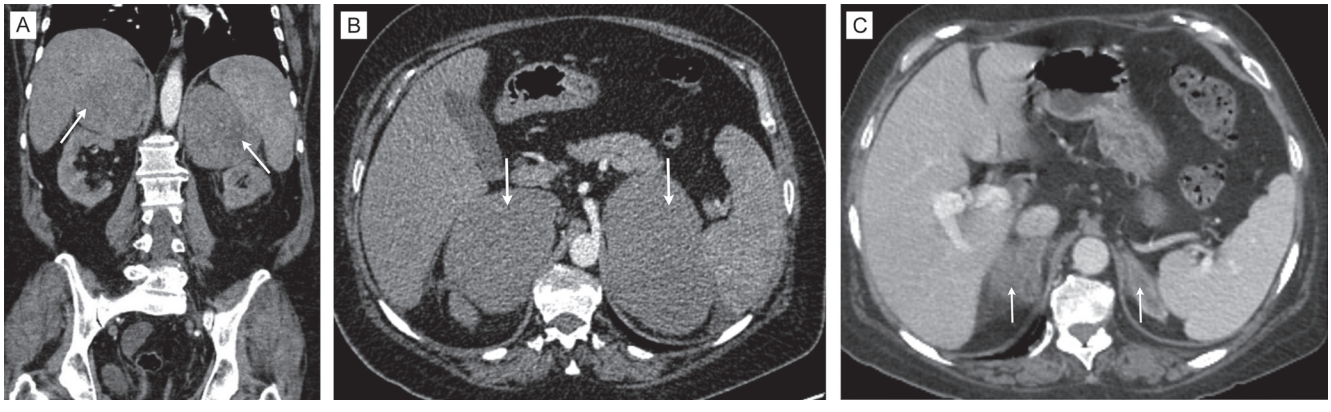


Figure 1. Baseline coronal (Panel A) and axial (Panel B) computed tomography (CT) demonstrates bilateral adrenal masses, 98 ×126 mm on left, unenhanced Hounsfield units (HU) of 23-40, and 86 × 118 mm on right, unenhanced HU of 35-51. A follow-up contrast-enhanced CT scan after 3 cycles of chemotherapy demonstrates a decrease in bilateral adrenal masses, now measuring 55 ×36 mm on the left and 47 × 38 mm on the right (Panel C).

of hyponatremia.

Acenocoumarol was discontinued and further replaced with heparin, as she had several bouts of atrial fibrillation. Blood products were administered with improvement of coagulability and anemia. Following this diagnosis, she initiated the first R-CHOP (Rituximab-Cyclophosphamide, Doxorubicin Hydrochloride, Vincristine, Prednisone) cycle of chemotherapy.

Outcome and follow-up

Following 76 days after initial diagnosis, the patient has completed the third cycle of chemotherapy, and a follow-up CT scan showed a significant decrease in the size of both adrenal masses, measuring 55 ×36 mm on the left and 47 × 38 mm on the right at that time (Fig. 1C). Follow up FDG-PET-CT scan revealed minimum FDG uptake in both adrenal masses. At the most recent follow up (123 days post-diagnosis) the patient has completed the fifth cycle of chemotherapy and eventually died due to irreversible cerebral neurologic complications.

DISCUSSION

We report a case of a patient who presented with primary adrenal insufficiency and an impending adrenal crisis due to primary bilateral adrenal lymphoma.

Primary adrenal lymphoma is a very rare adrenal malignancy reported to occur in 0.2% of all adrenal masses in a population-based study (1), and in 2-5% of all adrenal malignancies (2, 3). Bilateral involvement of adrenal glands in patients with primary adrenal lymphoma was reported in up to 30% of cases (2). In a study that included 249 patients with bilateral adrenal malignancies, primary adrenal insufficiency was reported in 12.4% (2). Notably, this was likely an underestimate as few patients had assessment for cortisol deficiency. Other authors suggest that adrenal insufficiency is rather common in bilateral PAL (4).Primary adrenal insufficiency occurs when >90% of adrenal cortex is destroyed. Most commonly, primary adrenal insufficiency is caused by the autoimmune adrenalitis, and only rarely by bilateral adrenal hemorrhage, or infiltrative disorders, such as fungal infections, tuberculosis, granulomatous disorders, or malignancy (2, 5, 6). Risk of primary adrenal insufficiency was 20% when adrenal mass was > 4 cm in diameter, and higher when the etiology of adrenal malignancy was lymphoma or melanoma (27%) (2). Once established, primary adrenal insufficiency is usually permanent (6).

Our patient who was referred for the evaluation of large bilateral adrenal masses reported symptoms and signs of impending adrenal crisis, and subsequently confirmed to have primary adrenal insufficiency based on electrolyte abnormalities, low cortisol, and elevated ACTH. In retrospect, she had symptoms suggestive of primary adrenal insufficiency for 2 months, however, due to the non-specific nature of her presentation, the diagnosis was delayed. Treatment for primary adrenal insufficiency should be initiated in any patient with suggestive symptoms and signs without waiting for laboratory confirmation (7). Considering patient's presentation with bilateral adrenal masses and

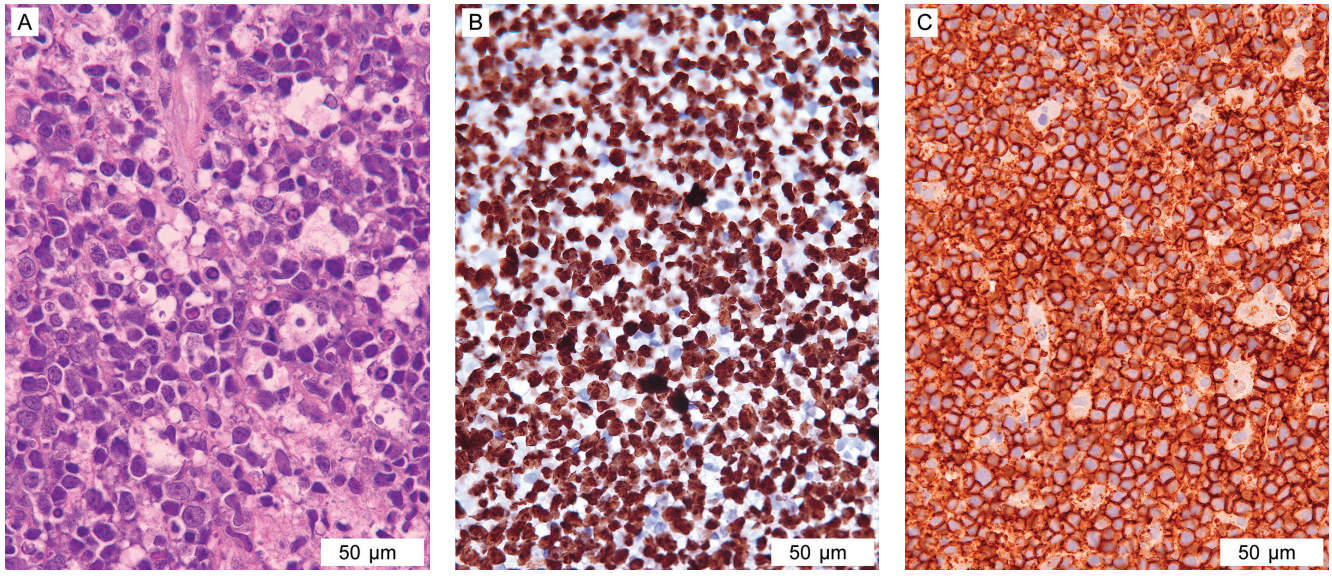


Figure 2. Biopsy of one of the adrenal masses demonstrated a diffuse large B-cell lymphoma (hematoxylin and eosin-stained x 600), Panel A, with CD 20 positivity, 400 x magnification, Panel B, and a positive Ki-67 immunohistochemistry, 400 x magnification, in ~90% of cells (Panel C).

symptoms of primary adrenal insufficiency, we initiated glucocorticoid and mineralocorticoid therapy prior to laboratory confirmation.

Differential diagnosis of our patient included bilateral adrenal hemorrhage, bilateral adrenal tuberculosis, fungal infection, granulomatous disease, or malignancy. In the absence of pertinent history or risk factors combined with work up that could be in this patient, making the noninvasive diagnosis was challenging. For example, our patient presented with severe anemia raising the possibility of adrenal hemorrhage, however, she did not have a known coagulopathy, and imaging findings were not consistent with blood products on CT. Bilateral adrenal metastases were considered unlikely as patient did not have a known extra-adrenal malignancy. Adrenocortical carcinoma was considered unlikely due to bilateral presentation, which is extremely unlikely with this etiology (8). Bilateral pheochromocytomas do not present with primary adrenal insufficiency, and frequently occur in patients with a known genetic syndrome (9). Presentation with both primary adrenal insufficiency and bilateral adrenal masses due to tuberculous adrenalitis is uncommon but possible in developing countries, like Romania. The diagnosis is suggested by the presence of extra-adrenal involvement, which our patient didn't have.

After stabilizing our patient and excluding pheochromocytoma, we proceeded with adrenal biopsy that confirmed the diagnosis. Adrenal biopsy was reported to have a 96-98% sensitivity and specificity for diagnosis of adrenal malignancy, and is recommended only in patients with indeterminate adrenal mass, after excluding adrenocortical carcinoma and pheochromocytoma, and when management depends on histology results (10). Our patient was treated with R-CHOP chemotherapy, with good response after 3 cycles.

In conclusion, we presented a case of large bilateral lymphomas diagnosed after presenting with primary adrenal insufficiency. Bilateral infiltrative adrenal malignancy is a very rare cause of primary adrenal insufficiency, and a high level of suspicion is needed for appropriate diagnosis and management.

Learning points:

- Primary adrenal insufficiency should be considered in patients presenting with bilateral adrenal masses.

- Although primary adrenal lymphoma is a very rare adrenal malignancy it should be considered in patients presenting with bilateral rapidly growing adrenal tumors and primary adrenal insufficiency.

- Adrenal biopsy is recommended to make a histological diagnosis after excluding adrenocortical carcinoma and pheochromocytoma

- The treatment of adrenal crisis should not be delayed until the hormonal and the etiologic diagnosis are made.

Conflict of interest

The authors declare that they have no conflict of interest.

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Informed Patient Consent for Publication

Signed informed consent obtained directly from the patient's relatives or guardians.

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